

MATH203 – APPLIED LINEAR ALGEBRA
HOMEWORK 3

Your Name: _____

Your Student Number: _____

Your Tutorial TA: _____

Instructions:

- (1) Show your work and justify your steps on every question but do not include extraneous information.
- (2) Please clearly indicate your name, as it is required to identify the recipient of the homework grade. Include the name of your tutorial TA so we can forward your submission to the appropriate grader. This also enables your TA to better understand the strengths and weaknesses of students in their section. Your student number is required for record-keeping and to streamline the grading process. Missing the student number may lead to delays in grading or misattribution of your work.
- (3) You may use any resources you like, but you must write up the solutions on your own. Plagiarism will not be tolerated and will be penalized according to the university's academic integrity policies. AI generated content is considered plagiarism.

1. (10 points) Answer each of the following True or False questions. If you answer False, give a counterexample.

1.1) $\det(A + B) = \det(A) + \det(B)$ for all $n \times n$ matrices A, B .

Solution:False. Let $A = B = I_2$. Then $\det(A) = \det(B) = 1$, but $\det(A + B) = \det(2I_2) = 4 \neq 2 = \det(A) + \det(B)$.

1.2) If A is a 3×3 matrix with $\det(A) = 2$, then $\det(3A) = 54$.

Solution:True. For an $n \times n$ matrix, $\det(cA) = c^n \det(A)$. Here $n = 3$ and $c = 3$, so $\det(3A) = 3^3 \cdot 2 = 54$.

1.3) $\det(AB) = \det(A) \det(B)$ for all $n \times n$ matrices A, B .

Solution: True. This is the product formula for determinants.

1.4) If A is a square matrix with two identical rows, then $\det(A) = 0$.

Solution: True. Swapping the two identical rows leaves the matrix unchanged but changes the sign of the determinant: $\det(A) = -\det(A)$, so $\det(A) = 0$.

1.5) If A is invertible, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

Solution: True. From $AA^{-1} = I$, taking determinants: $\det(A)\det(A^{-1}) = \det(I) = 1$, so $\det(A^{-1}) = 1/\det(A)$.

2. (25 points) Interpolation and the Vandermonde determinant.

2.1) Let a, b, c be real numbers, and let

$$V = \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}.$$

Show that $\det(V) = (b-a)(c-a)(c-b)$.

Solution: $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\det(V) = \det \begin{pmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{pmatrix} = \begin{vmatrix} b-a & b^2-a^2 \\ c-a & c^2-a^2 \end{vmatrix}.$$

Factor $(b-a)$ from row 1 and $(c-a)$ from row 2:

$$= (b-a)(c-a) \begin{vmatrix} 1 & b+a \\ 1 & c+a \end{vmatrix} = (b-a)(c-a)(c-b).$$

2.2) Let x_1, x_2, x_3 be distinct real numbers. We seek the polynomial $y = P(x) = a_0 + a_1x + a_2x^2$ whose graph passes through the three points

$$(x, y) = (x_1, y_1), (x_2, y_2), (x_3, y_3),$$

i.e. $P(x_i) = y_i$ for $i = 1, 2, 3$. In matrix form this is $V\mathbf{a} = \mathbf{y}$ with

$$\underbrace{\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix}}_V \underbrace{\begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}}_{\mathbf{a}} = \underbrace{\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}}_{\mathbf{y}}.$$

By **Cramer's rule**, $a_k = \det(V_k)/\det(V)$, where V_k is obtained from V by replacing *column* $k+1$ with $\mathbf{y}$:

$$V_0 = \underbrace{\begin{pmatrix} y_1 & x_1 & x_1^2 \\ y_2 & x_2 & x_2^2 \\ y_3 & x_3 & x_3^2 \end{pmatrix}}_{\text{col } 1 \leftarrow \mathbf{y}}, \quad V_1 = \underbrace{\begin{pmatrix} 1 & y_1 & x_1^2 \\ 1 & y_2 & x_2^2 \\ 1 & y_3 & x_3^2 \end{pmatrix}}_{\text{col } 2 \leftarrow \mathbf{y}}, \quad V_2 = \underbrace{\begin{pmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{pmatrix}}_{\text{col } 3 \leftarrow \mathbf{y}}.$$

Show that $P(x)$ can be rewritten as

$$P(x) = \sum_{i=1}^3 y_i \cdot \frac{\det(\tilde{V}_i(x))}{\det(V)},$$

where $\tilde{V}_i(x)$ is obtained from V by replacing *row* i with $(1, x, x^2)$:

$$\tilde{V}_1(x) = \underbrace{\begin{pmatrix} 1 & x & x^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix}}_{\text{row } 1 \leftarrow (1, x, x^2)}, \quad \tilde{V}_2(x) = \underbrace{\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x & x^2 \\ 1 & x_3 & x_3^2 \end{pmatrix}}_{\text{row } 2 \leftarrow (1, x, x^2)}, \quad \tilde{V}_3(x) = \underbrace{\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x & x^2 \end{pmatrix}}_{\text{row } 3 \leftarrow (1, x, x^2)}.$$

Then, using the Vandermonde formula from (a), show that

$$\frac{\det(\tilde{V}_1(x))}{\det(V)} = \frac{(x-x_2)(x-x_3)}{(x_1-x_2)(x_1-x_3)} = L_1(x).$$

Hint: Laplace-expand $\det(V_k)$ along its $\mathbf{y}$ -column, substitute into $P(x) = \sum_k a_k x^k$, and swap the order of summation.

Solution: Step 1 (Cramer's rule). $a_k = \det(V_k)/\det(V)$, with V_k the column-replaced matrix displayed above.

Step 2 (Laplace expansion of $\det(V_k)$ along column $k+1$). The $(i, k+1)$ -entry of V_k is y_i ; deleting row i and column $k+1$ of V_k erases the replaced column, so the minor matches that of V and $C_{i,k+1}(V_k) = C_{i,k+1}(V)$. Hence

$$\det(V_k) = \sum_{i=1}^3 y_i \cdot \underbrace{C_{i,k+1}(V)}_{\text{depends only on } V}.$$

For example, $k = 0$ (expansion of $\det(V_0)$ along column 1) gives $\det(V_0) = y_1 C_{1,1}(V) + y_2 C_{2,1}(V) + y_3 C_{3,1}(V)$, where each cofactor is (up to sign) the minor obtained by deleting the indicated row and column

of V :

$$\begin{aligned}
C_{1,1}(V) &= + \det \underbrace{\begin{pmatrix} x_2 & x_2^2 \\ x_3 & x_3^2 \end{pmatrix}}_{\text{delete row 1, col 1}}, \\
C_{2,1}(V) &= - \det \underbrace{\begin{pmatrix} x_1 & x_1^2 \\ x_3 & x_3^2 \end{pmatrix}}_{\text{delete row 2, col 1}}, \\
C_{3,1}(V) &= + \det \underbrace{\begin{pmatrix} x_1 & x_1^2 \\ x_2 & x_2^2 \end{pmatrix}}_{\text{delete row 3, col 1}}.
\end{aligned}$$

Step 3 (substitute into $P(x)$ and swap sums).

$$P(x) = \sum_{k=0}^2 a_k x^k = \frac{1}{\det V} \sum_{k=0}^2 x^k \det(V_k) = \frac{1}{\det V} \sum_{k=0}^2 x^k \sum_{i=1}^3 y_i C_{i,k+1}(V) = \frac{1}{\det V} \sum_{i=1}^3 y_i \underbrace{\sum_{k=0}^2 x^k C_{i,k+1}(V)}_{(\star)_i}.$$

Step 4 (recognize $(\star)_i$ as $\det(\tilde{V}_i(x))$). By construction, $\tilde{V}_i(x)$ agrees with V in every row except row i , which is $(1, x, x^2)$. Hence the cofactors along row i of $\tilde{V}_i(x)$ coincide with those of V : $C_{i,j}(\tilde{V}_i(x)) = C_{i,j}(V)$. Laplace-expanding $\tilde{V}_i(x)$ along row i gives

$$\det(\tilde{V}_i(x)) = 1 \cdot C_{i,1}(V) + x \cdot C_{i,2}(V) + x^2 \cdot C_{i,3}(V) = (\star)_i.$$

For example, for $i = 1$:

$$\det(\tilde{V}_1(x)) = 1 \cdot \underbrace{\det \begin{pmatrix} x_2 & x_2^2 \\ x_3 & x_3^2 \end{pmatrix}}_{C_{1,1}(V)} - x \cdot \underbrace{\det \begin{pmatrix} 1 & x_2^2 \\ 1 & x_3^2 \end{pmatrix}}_{-C_{1,2}(V)} + x^2 \cdot \underbrace{\det \begin{pmatrix} 1 & x_2 \\ 1 & x_3 \end{pmatrix}}_{C_{1,3}(V)}.$$

Combining steps 1–4:

$$\boxed{P(x) = \sum_{i=1}^3 y_i \cdot \frac{\det(\tilde{V}_i(x))}{\det(V)}}.$$

Step 5 (evaluate via the Vandermonde formula). The matrix $\tilde{V}_1(x)$ is itself Vandermonde, on the nodes x, x_2, x_3 :

$$\tilde{V}_1(x) = \underbrace{\begin{pmatrix} 1 & x & x^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix}}_{\text{Vandermonde on } x, x_2, x_3}.$$

By (a),

$$\det(\tilde{V}_1(x)) = (x_2 - x)(x_3 - x)(x_3 - x_2), \quad \det(V) = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2).$$

Cancelling $(x_3 - x_2)$ and flipping signs in both numerator factors and both denominator factors:

$$\frac{\det(\tilde{V}_1(x))}{\det(V)} = \frac{(x_2 - x)(x_3 - x)}{(x_2 - x_1)(x_3 - x_1)} = \frac{(x - x_2)(x - x_3)}{(x_1 - x_2)(x_1 - x_3)} = L_1(x). \quad \checkmark$$

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2.3) Find the polynomial $P(x) = a_0 + a_1x + a_2x^2$ of degree ≤ 2 passing through $(0, 1), (1, 2), (3, 10)$.

Solution: The interpolation system $V\mathbf{a} = \mathbf{y}$ is

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 10 \end{pmatrix}, \quad \det(V) = (1-0)(3-0)(3-1) = 6.$$

By Cramer's rule, $a_k = \det(V_k) / \det(V)$ with V_k obtained from V by replacing column $k+1$ with $\mathbf{y}$:

$$\det(V_0) = \det \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 10 & 3 & 9 \end{pmatrix} = 6, \quad \det(V_1) = \det \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 10 & 9 \end{pmatrix} = 0, \quad \det(V_2) = \det \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 3 & 10 \end{pmatrix} = 6.$$

So $a_0 = 1$, $a_1 = 0$, $a_2 = 1$, giving $\boxed{P(x) = x^2 + 1}$. Check: $P(0) = 1, P(1) = 2, P(3) = 10$.

2.4) Let $x_1, \dots, x_n$ be n distinct real numbers. Show that for any $y_1, \dots, y_n$, there exists a **unique** polynomial $P(x)$ of degree $\leq n-1$ with $P(x_k) = y_k$ for all k . (You may use the formula $\det(V_n) = \prod_{i < j} (x_j - x_i)$.)

Solution: Write $P(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1}$. The conditions $P(x_k) = y_k$ give $V_n\mathbf{a} = \mathbf{y}$. Since the x_k are distinct, every factor $(x_j - x_i)$ is nonzero, so $\det(V_n) \neq 0$ and V_n is invertible. Hence $\mathbf{a}$ (and therefore P) is uniquely determined.

3. (15 points) Let A be an $n \times m$ matrix and B be an $m \times n$ matrix.

3.1) Show that $\det(I_n - AB) = \det(I_m - BA)$. (Note: $I_n - AB$ and $I_m - BA$ have **different sizes** in general.)

*Hint: apply two different block-triangularizations to the $(n+m) \times (n+m)$ block matrix $\begin{pmatrix} I_n & A \\ B & I_m \end{pmatrix}$. You may use $\det \begin{pmatrix} X & * \\ 0 & Y \end{pmatrix} = \det(X) \det(Y)$.*

Solution: Let $M = \begin{pmatrix} I_n & A \\ B & I_m \end{pmatrix}$. Multiplying on the left by two different block-unit-triangular matrices (each has $\det = 1$):

Elimination 1: $\begin{pmatrix} I_n & -A \\ 0 & I_m \end{pmatrix} M = \begin{pmatrix} I_n - AB & 0 \\ B & I_m \end{pmatrix}$. So $\det(M) = \det(I_n - AB) \cdot \det(I_m) = \det(I_n - AB)$.

Elimination 2: $\begin{pmatrix} I_n & 0 \\ -B & I_m \end{pmatrix} M = \begin{pmatrix} I_n & A \\ 0 & I_m - BA \end{pmatrix}$. So $\det(M) = \det(I_n) \cdot \det(I_m - BA) = \det(I_m - BA)$.

Combining: $\det(I_n - AB) = \det(M) = \det(I_m - BA)$.

3.2) Compute the determinant of

$$M = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

Solution: Let $\mathbf{1} = (1, 1, 1, 1, 1)^T \in \mathbb{R}^5$. Then the all-ones 5×5 matrix equals $\mathbf{1}\mathbf{1}^T$, and

$$M = \underbrace{\mathbf{1}\mathbf{1}^T}_{\text{all-ones } 5 \times 5} - I_5 = -(I_5 - \mathbf{1}\mathbf{1}^T),$$

so

$$\det(M) = (-1)^5 \det(I_5 - \mathbf{1}\mathbf{1}^T) = -\det(I_5 - \mathbf{1}\mathbf{1}^T).$$

Now apply (a) with $A = \underbrace{\mathbf{1}}_{5 \times 1}$ and $B = \underbrace{\mathbf{1}^T}_{1 \times 5}$:

$$\underbrace{AB}_{5 \times 5} = \mathbf{1}\mathbf{1}^T, \quad \underbrace{BA}_{1 \times 1} = \mathbf{1}^T \mathbf{1} = 5.$$

By the identity,

$$\det(I_5 - \mathbf{1}\mathbf{1}^T) = \det(I_1 - 5) = \det(-4) = -4.$$

Therefore $\det(M) = -(-4) = \boxed{4}$.

3.3) Show that if A, B are $n \times n$ matrices with $AB = I_n$, then $BA = I_n$.

Solution: Let $P = BA$. Since $AB = I_n$,

$$P^2 = (BA)(BA) = B(AB)A = BI_nA = BA = P.$$

Thus P is a projection. Therefore $I_n - P = I_n - BA$ is also a projection:

$$(I_n - P)^2 = I_n - 2P + P^2 = I_n - P.$$

Now use $\text{tr}(BA) = \text{tr}(AB)$. We get

$$\text{tr}(I_n - BA) = \text{tr}(I_n) - \text{tr}(BA) = n - \text{tr}(AB) = n - \text{tr}(I_n) = 0.$$

By the lecture result for projection matrices, $\text{rank}(Q) = \text{tr}(Q)$ for any projection matrix Q . Hence

$$\text{rank}(I_n - BA) = \text{tr}(I_n - BA) = 0.$$

So $I_n - BA = 0$, and therefore $BA = I_n$.

TA note: This problem is not strongly related to the major problem of this exercise. No matter what proof a student gives, as long as the proof is correct, please give full credit.

4. (15 points) Consider the data

x_i	-1	0	1	2
y_i	1	5	1	9

Find the polynomial $P(x) = a_0 + a_1x + a_2x^2$ of degree ≤ 2 minimizing

$$\sum_{i=1}^4 (P(x_i) - y_i)^2.$$

Solution: Let B be the 4×3 matrix whose (i, j) -entry is x_i^{j-1} (value of the monomial x^{j-1} at x_i), and let $\mathbf{b} = (y_1, y_2, y_3, y_4)^T$. Then $B\hat{\mathbf{a}}$, with $\hat{\mathbf{a}} = (a_0, a_1, a_2)^T$, is the vector of predicted values $(P(x_1), \dots, P(x_4))^T$, and the objective equals $\|B\hat{\mathbf{a}} - \mathbf{b}\|^2$. Minimizing this is the orthogonal projection of $\mathbf{b}$ onto $\text{Col}(B)$, whose coefficient vector is $\hat{\mathbf{a}} = (B^T B)^{-1} B^T \mathbf{b}$.

$$B = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}, \quad B^T B = \begin{pmatrix} 4 & 2 & 6 \\ 2 & 6 & 8 \\ 6 & 8 & 18 \end{pmatrix}, \quad B^T \mathbf{b} = \begin{pmatrix} 16 \\ 18 \\ 38 \end{pmatrix}.$$

Applying row operations to both sides of $\hat{\mathbf{a}} = (B^T B)^{-1} B^T \mathbf{b}$ simultaneously:

$R_2 \rightarrow 2R_2 - R_1, R_3 \rightarrow 2R_3 - 3R_1$:

$$\hat{\mathbf{a}} = \begin{pmatrix} 4 & 2 & 6 \\ 0 & 10 & 10 \\ 0 & 10 & 18 \end{pmatrix}^{-1} \begin{pmatrix} 16 \\ 20 \\ 28 \end{pmatrix}.$$

$R_3 \rightarrow R_3 - R_2; R_3 \rightarrow \frac{1}{8}R_3$:

$$\hat{\mathbf{a}} = \begin{pmatrix} 4 & 2 & 6 \\ 0 & 10 & 10 \\ 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 16 \\ 20 \\ 1 \end{pmatrix}.$$

$R_1 \rightarrow R_1 - 6R_3, R_2 \rightarrow R_2 - 10R_3; R_2 \rightarrow \frac{1}{10}R_2$:

$$\hat{\mathbf{a}} = \begin{pmatrix} 4 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 10 \\ 1 \\ 1 \end{pmatrix}.$$

$R_1 \rightarrow R_1 - 2R_2; R_1 \rightarrow \frac{1}{4}R_1$:

$$\hat{\mathbf{a}} = I^{-1} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

So $\boxed{P(x) = 2 + x + x^2}$. Check: residual $\mathbf{e} = \mathbf{b} - B\hat{\mathbf{a}} = (-1, 3, -3, 1)^T$ satisfies $B^T \mathbf{e} = \mathbf{0}$, confirming $B\hat{\mathbf{a}}$ is the projection. 5

5. (15 points) **Euclidean division for matrix polynomials.**

A **matrix polynomial** has the form

$$F(\lambda) = A_p \lambda^p + A_{p-1} \lambda^{p-1} + \cdots + A_1 \lambda + A_0,$$

where each A_i is an $n \times n$ matrix and $A_p \neq 0$. It is called **monic** if its leading coefficient equals the identity: $A_p = I$.

Prove: for any matrix polynomial F and any monic matrix polynomial d of degree k , there exist **unique** matrix polynomials Q (quotient) and R (remainder) with

$$F(\lambda) = Q(\lambda) d(\lambda) + R(\lambda), \quad \deg R < k.$$

5.1) **Existence.** Prove by induction on $p = \deg F$.

Hint: if $p \geq k$, consider $\tilde{F} = F - A_p \lambda^{p-k} d$ and show $\deg \tilde{F} < p$.

Solution:Base. If $p < k$: take $Q = 0$, $R = F$. Then $F = 0 \cdot d + F$ with $\deg R < k$.

Step. Assume the result for all degrees $< p$, and let $p \geq k$. Define

$$\tilde{F}(\lambda) = F(\lambda) - A_p \lambda^{p-k} d(\lambda).$$

The leading term of $A_p \lambda^{p-k} d(\lambda)$ is $A_p \cdot I \cdot \lambda^p = A_p \lambda^p$ (using d monic), which cancels the leading term of F . So $\deg \tilde{F} < p$.

By induction, $\tilde{F} = \tilde{Q} d + R$ with $\deg R < k$. Then

$$F = A_p \lambda^{p-k} d + \tilde{F} = \underbrace{(A_p \lambda^{p-k} + \tilde{Q})}_Q d + R,$$

with $\deg R < k$ as required.

5.2) **Uniqueness.** Prove Q and R are unique.

Hint: if $F = Q_1 d + R_1 = Q_2 d + R_2$, look at $(Q_1 - Q_2) d$.

Solution: Suppose $F = Q_1 d + R_1 = Q_2 d + R_2$ with $\deg R_1, \deg R_2 < k$. Then

$$(Q_1 - Q_2) d = R_2 - R_1,$$

and the right-hand side has degree $< k$.

If $Q_1 \neq Q_2$, let $C_j \lambda^j$ be the leading term of $Q_1 - Q_2$ with $C_j \neq 0$. Since d is monic, the leading term of $(Q_1 - Q_2) d$ is $C_j \lambda^{j+k}$, which is nonzero and has degree $\geq k$ — contradicting $\deg(R_2 - R_1) < k$. So $Q_1 = Q_2$, and hence $R_1 = R_2$.

5.3) **Application.** Let M be a fixed $n \times n$ matrix and take the monic linear divisor $d(\lambda) = \lambda I - M$. For any matrix polynomial $F(\lambda) = \sum_{p \geq 0} A_p \lambda^p$, the division

$$F(\lambda) = Q(\lambda) (\lambda I - M) + R$$

has $\deg R < 1$, so R is a constant $n \times n$ matrix. Show that

$$R = \sum_{p \geq 0} A_p M^p,$$

i.e., R is obtained from $F(\lambda)$ by substituting $\lambda = M$ on the *right* of each coefficient A_p .

Remark: applying this to $F(\lambda) = \chi_M(\lambda) I$ (the characteristic polynomial of M times the identity matrix) recovers the Cayley–Hamilton theorem.

Solution: Define the right-substitution map

$$E\left(\sum_{p \geq 0} A_p \lambda^p\right) = \sum_{p \geq 0} A_p M^p.$$

We apply E to both sides of

$$F(\lambda) = Q(\lambda)(\lambda I - M) + R.$$

Write $Q(\lambda) = \sum_{j \geq 0} B_j \lambda^j$. Then

$$Q(\lambda)(\lambda I - M) = \sum_{j \geq 0} B_j \lambda^{j+1} - \sum_{j \geq 0} B_j \lambda^j M.$$

After right-substitution,

$$E(Q(\lambda)(\lambda I - M)) = \sum_{j \geq 0} B_j M^{j+1} - \sum_{j \geq 0} B_j M^j M = 0.$$

Also $E(R) = R$ because R is constant. Therefore

$$\sum_{p \geq 0} A_p M^p = E(F) = E(Q(\lambda)(\lambda I - M)) + E(R) = R.$$

So the remainder is exactly

$$R = \sum_{p \geq 0} A_p M^p.$$

6. (15 points) Rank of the adjugate. Let A be an $n \times n$ matrix. The **adjugate** A^* is the matrix whose entries are the (signed) $(n-1) \times (n-1)$ minors of A ; it satisfies the identity

$$A A^* = \det(A) I_n. \quad (\star)$$

You may use $(\star)$ without proof. Prove the following three statements.

6.1) If $\text{rank}(A) \leq n-2$, then $A^* = 0$.

Solution: Every entry of A^* is $\pm \det$ of an $(n-1) \times (n-1)$ submatrix of A . If $\text{rank}(A) \leq n-2$, then every such submatrix has $\text{rank} \leq n-2 < n-1$, hence determinant 0. So $A^* = 0$.

6.2) If $\text{rank}(A) = n$, then $\text{rank}(A^*) = n$.

Solution: Then $\det(A) \neq 0$, so $(\star)$ gives $A^* = \det(A) A^{-1}$ — a nonzero scalar multiple of an invertible matrix, hence invertible. So $\text{rank}(A^*) = n$.

6.3) If $\text{rank}(A) = n-1$, then $\text{rank}(A^*) = 1$.

Hint: use $(\star)$ for the upper bound; find a nonzero minor for the lower bound.

Solution: Upper bound. $\det(A) = 0$, so $(\star)$ becomes $A A^* = 0$. Every column of A^* lies in $\text{Null}(A)$, which has dimension $n - (n-1) = 1$. So all columns are multiples of a single vector, giving $\text{rank}(A^*) \leq 1$.

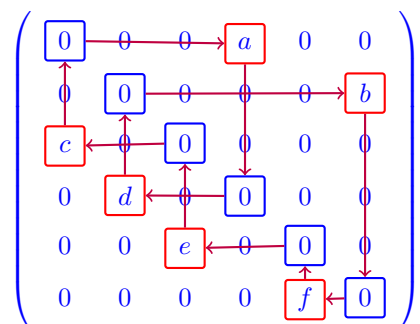
Lower bound. Since $\text{rank}(A) = n-1$, some $(n-1) \times (n-1)$ submatrix of A has full rank, i.e., some minor is nonzero. So $A^* \neq 0$, giving $\text{rank}(A^*) \geq 1$.

Combining: $\text{rank}(A^*) = 1$.

7. (10 points) Compute the determinant of

$$M = \begin{pmatrix} 0 & 0 & 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & b \\ c & 0 & 0 & 0 & 0 & 0 \\ 0 & d & 0 & 0 & 0 & 0 \\ 0 & 0 & e & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & f & 0 \end{pmatrix}.$$

Solution: Use the zigzag method from the determinant lecture. Box the nonzero entries in red, and mark the diagonal squares in blue:



Each row and each column has exactly one boxed entry. Therefore

$$\det(M) = abcdef \cdot \det(S),$$

where S is the switching matrix with 1 in the same boxed positions:

$$S = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

The arrows show one zigzag loop, so $m = 1$. By the switching-matrix formula from the lecture,

$$\det(S) = (-1)^{n+m} = (-1)^{6+1} = -1.$$

Hence

$$\boxed{\det(M) = abcdef \cdot (-1) = -abcdef}.$$

0

8. (10 points) For $n \geq 1$, let

$$T_n = \begin{pmatrix} -1 & 2 & 0 & \cdots & \cdots & 0 \\ -1 & 1 & 2 & \ddots & & \vdots \\ 0 & -1 & 1 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ \vdots & & & \ddots & -1 & 1 & 2 \\ 0 & \cdots & \cdots & 0 & -1 & 1 \end{pmatrix}.$$

Compute $\det(T_n)$.

Solution: Let $d_n = \det(T_n)$. Directly, $d_1 = -1$ and $d_2 = (-1)(1) - (2)(-1) = 1$.

Write the first cross-filling step as a standard rank-1 decomposition. Use the pivot $(1, 1) = -1$:

$$\underbrace{\begin{pmatrix} -1 & 2 & 0 & \cdots & 0 \\ -1 & 1 & 2 & \ddots & \vdots \\ 0 & -1 & 1 & 2 & 0 \\ \vdots & \ddots & \ddots & \ddots & 2 \\ 0 & \cdots & 0 & -1 & 1 \end{pmatrix}}_{T_n} = \frac{1}{-1} \underbrace{\begin{pmatrix} -1 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{R_1} \underbrace{(-1 \ 2 \ 0 \ \cdots \ 0)}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 2 & \ddots & \vdots \\ 0 & -1 & 1 & 2 & 0 \\ \vdots & \ddots & \ddots & \ddots & 2 \\ 0 & \cdots & 0 & -1 & 1 \end{pmatrix}}_{T_n - R_1}.$$

Equivalently,

$$T_n = R_1 + \begin{pmatrix} 0 & 0 \\ 0 & T_{n-1} \end{pmatrix}.$$

So the cross-center is -1 , and the remaining block is exactly T_{n-1} .

By the determinant rule for cross-filling,

$$d_n = (-1) \cdot d_{n-1}.$$

Iterating from $d_1 = -1$: $\boxed{\det(T_n) = (-1)^n} \cdot 0$